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# Designing an agentic AI workflow for structured information extraction from scientific text: A case study on atomic layer deposition of ZnO and IGZO

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**Note:** This paper is part of the Special Topic on Atomic Layer Deposition (ALD).
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## ABSTRACT

The rapid growth of scientific literature in materials sciences and other domains presents a growing challenge where critical process knowledge remains embedded in unstructured text, limiting its reuse for data-driven discovery. Agentic workflows, which integrate large language models (LLMs) with different tools, such as application programming interfaces (APIs), and rule-based functions, have emerged as a promising paradigm to overcome these limitations. In this work, we introduce *SciKGExtract*, an agentic artificial intelligence (AI) workflow for structured knowledge extraction from scientific publications. The workflow combines LLM-based contextual understanding with deterministic tools for schema validation, data cleaning, and normalization, leveraging authoritative external repositories such as PubChem to ensure accuracy and semantic consistency. Applied to atomic layer deposition (ALD) literature from the AtomicLimits database, *SciKGExtract* produces a structured dataset capturing experimental parameters and material properties across test cases of zinc oxide (ZnO) and indium-gallium-zinc oxide (*IGZO*). The extracted data reveal dominant ALD methods and chemistries, variability in reported growth-per-cycle and temperature ranges, and the need for richer metadata to enable reliable cross-study comparisons. These results demonstrate how structured extraction can transform unstructured literature into AI-ready knowledge, accelerating validation, benchmarking, and discovery in emerging materials systems.

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## I. INTRODUCTION

The accelerating growth of scientific literature across science, in general, and materials science, in particular,[1–3] has created both an opportunity and a significant challenge for the community. Recent advances in experimental techniques and computational methods have led to an increase in published articles. While this continuous expanding body of scientific literature contains a wealth of domain knowledge and insights, its huge volume has made it difficult for the researchers to identify, query, and compare relevant finding across different studies. Due to this, important experimental details and findings are buried within these large collections of publications, limiting their accessibility and reuse.

Structured knowledge refers to the information that is organized in a standardized, machine-readable format, where properties, such as materials, process conditions, and experimental details and their values, follow a consistent set of guidelines. Common structured representation includes formats like JSON or

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tabular data, where each property has a defined meaning and structure. Structured scientific literature offers several advantages over the traditional unstructured formats like text. First, it allows an efficient method for information retrieval and precise querying, allowing researchers to quickly locate key information without manually reviewing large number of publications. Second, it allows a systematic comparison across studies, making easier for the researchers to identify trends and inconsistencies within the data. Importantly, structured knowledge supports FAIR (Findable, Accessible, Interoperable and Reusable)[4] guiding principle for the scientific data and highlights the need for structured data infrastructures and automated extraction workflows that can transform the growing scientific literature into FAIR-compliant scientific knowledge.

Building any structured data representation relies on three key components. First, a schema (Fig. 1), which specifies the structure of data for a particular process or experiment. The schema specifies what information should be captured, such as materials, precursors, or temperature as well as their data type and constraints. Second, a shared vocabulary (often referred to as ontology in computer science) to provide consistent definitions for different domain-specific terms to ensure consistent scientific concepts across different studies. An example from material science domain is MatOnto,[5] which formalizes relationship between different material science concepts. Third, a knowledge graph (KG), which is a graph-based data representation that link entities (represented as nodes) based on their relationships (represented as edges). A KG allows information from multiple publications to be linked, compared, and analyzed and enables integration with existing databases.

Several community-driven initiatives demonstrate the benefits of having structured representation of scientific knowledge for effective access and reuse of research results. The first prominent initiative is Open Research Knowledge Graph (ORKG),[6] which represents core contributions of scholarly articles in a structured, machine-readable format. To capture different research findings, ORKG uses structured templates, similar to process schemas to describe and model key elements like methods and results, enabling direct comparison across multiple scientific publications. Another example, particularly in material science domain, is the Materials Knowledge Graph (MatKG),[7] which focuses on organizing entities such as materials, synthesis methods, and characterization techniques, for enabling data-driven materials discovery. Together, ORKG and MatKG illustrate the value of structured scholarly knowledge for large-scale analysis in material sciences.

Transforming a scientific publication into structured representation is a major challenge. Scientific articles are written for human readers and important details are often distributed across tables, figures, and text. Additionally, different researchers can use different scientific notations, units, and terminologies across publications, and as a result, manual curation of structured knowledge became time-consuming, error-prone, and difficult to scale as scientific literature continues to grow. Recent advances in artificial intelligence (AI) tools, specifically large language models (LLMs), offer a promising solution to these challenges. LLMs are large neural networks that are trained on large collection of text that can identify patterns and context in natural language. They have the ability to identify relevant entities (e.g., precursors, coreactants, and temperature), and their relationship within the text makes them ideal for extracting structured information from the scientific

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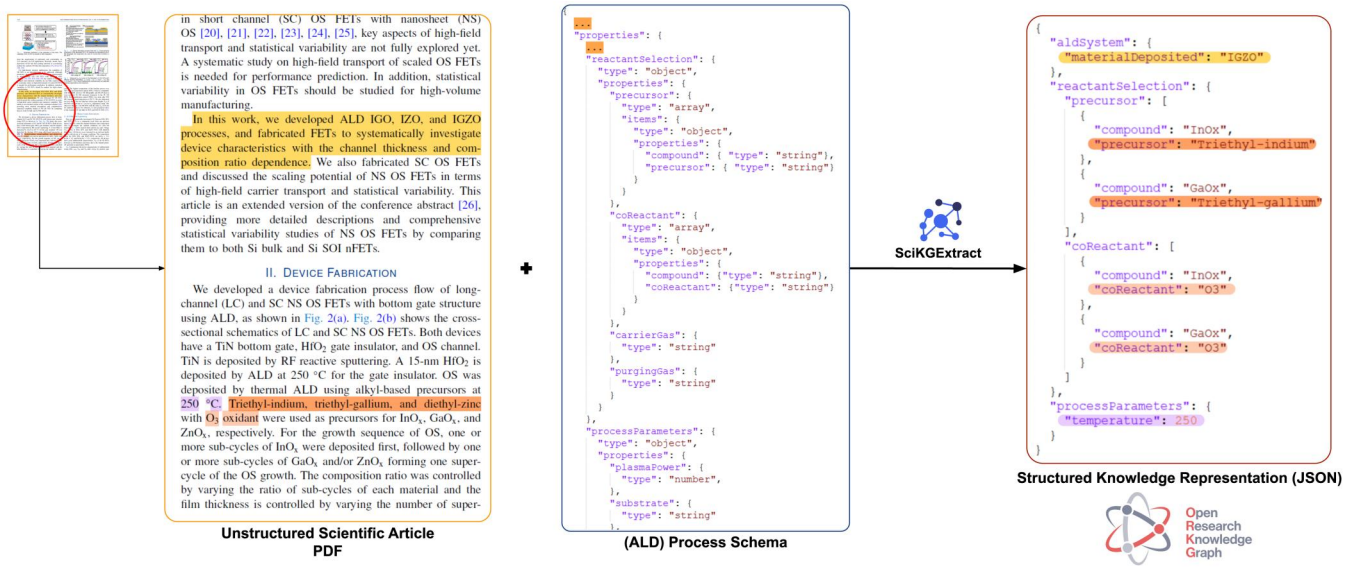



**FIG. 1.** Overview of schema-guided knowledge extraction from an unstructured scientific article. The system enables the schema-guided transformation of unstructured scientific literature, which is normalized, interoperable, and ready for integration into different scientific knowledge graphs like ORKG (Ref. 6). Reused with permission from Huang *et al.*, IEEE Trans. Electron Devices **71**, 7509–7515 (2024). Copyright 2024 IEEE, licensed under a Creative Commons License (Ref. 10).


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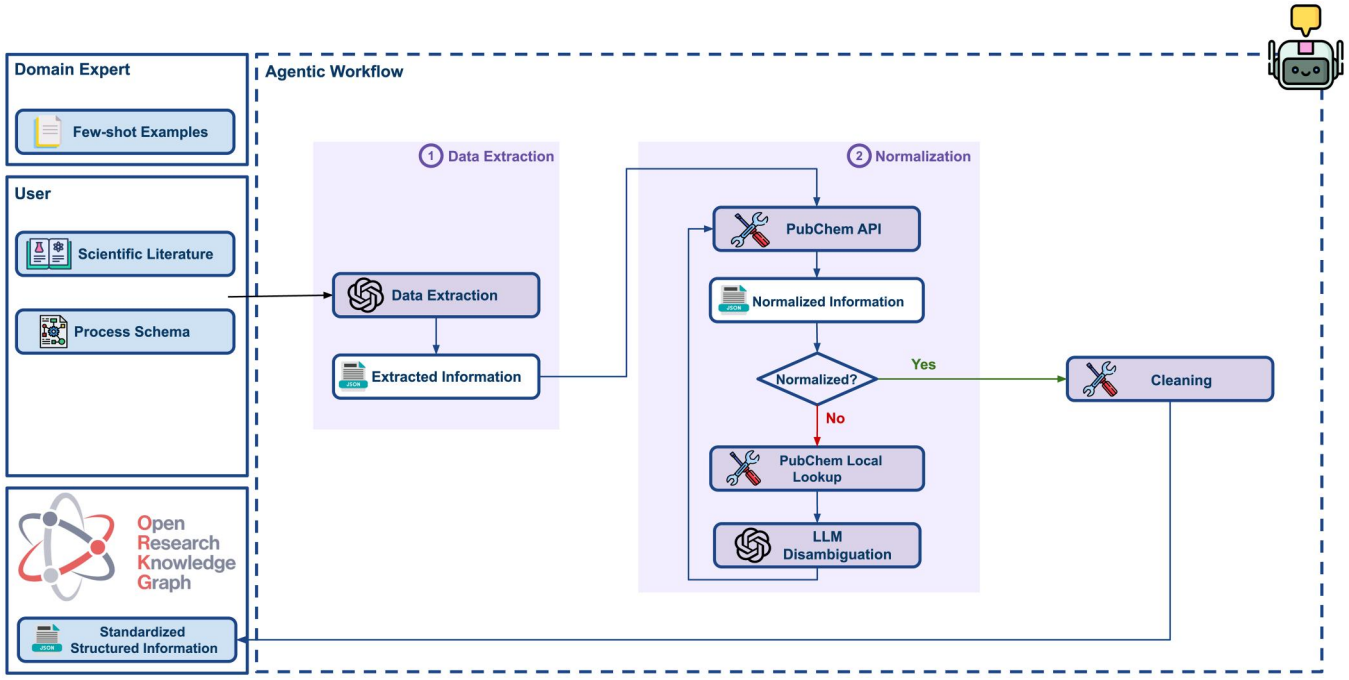



**FIG. 2.** Overview of the proposed *SciKGExtract* workflow for structured knowledge extraction. The workflow comprises structured data extraction, validation against the defined schema representation, normalization by grounding chemical compounds using PubChem data repository, and finally, cleaning of the normalized, schema-compliant information by removing empty, duplicate, or contextually invalid values.

articles. When combined with schema-guided extractions, where a predefined schema specify which properties to extract and how they should be organized, LLMs can transform on scale unstructured literature into consistent, machine-readable representations.

Building on these advances in AI, we propose *SciKGExtract* (Fig. 2), a modular and agentic workflow for structured knowledge extraction from scientific literature. An agentic workflow refers to a system that uses LLMs as a central reasoning unit that interacts with a collection of specialized tools, each responsible for a well-defined task. In *SciKGExtract*, LLMs perform schema-guided knowledge extraction from scientific documents, together with deterministic tools for handling tasks such as structure validation, normalization, and data cleaning. By combining the contextual understanding capabilities of LLMs together with rule-based tools and external scientific databases, *SciKGExtract* produces structured information that are both accurate and interoperable.

In this work, we used atomic layer deposition (ALD) process as a use case for evaluating structured knowledge extraction using *SciKGExtract* workflow. ALD is a vapor-phase thin-film deposition technique based on sequential, self limiting surface reactions, enabling precise control over film thickness and uniformity at the atomic scale.[8] A typical ALD process is characterized by recurring cycle of entities such as precursors, purge steps, and coreactant exposure at a certain deposition temperature and on particular substrate materials that are reported explicitly in the literature. As a result, ALD publications contain rich experimental descriptions that are well suited to schema-guided representations. We used AtomicLimits ALD database,[9] a curated community-driven resource that aggregates ALD-related scientific publications and provides manually annotated experimental metadata. Specifically, in this work, we focus on zinc oxide (ZnO) and indium-gallium-zinc-oxide (*IGZO*) ALD processes due to their relevance in semiconductor applications and optoelectronics.

The main contributions of this work are as follows:

- **Agentic workflow for structured knowledge extraction**: A schema-guided extraction workflow that integrates LLM reasoning with rule-based tools for validation, cleaning, and normalization for unifying scientific knowledge.
- **Empirical evaluation on atomic layer deposition (ALD) literature**: Validating both quantitatively and qualitatively the *SciKGExtract* workflow on experimental ALD publications from the AtomicLimits Database, together with curating structured ALD knowledge base for future comparisons.

The rest of the paper is organized as follows. Section II reviews the related work on LLM-based knowledge extraction and agentic workflows. Section III describes the *SciKGExtract* methodology, including schema-guided extraction, normalization, and cleaning steps. Section IV presents the experiments and results covering quantitative and qualitative evaluations on ALD literature and finally; Sec. V concludes the paper and outlines directions for future work.

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## II. RELATED WORK

In this section, we review two lines of work related to our study: existing workflows for LLM-based knowledge extraction and agentic-based approaches that combine LLMs with external tools for enhanced reliability and semantic structuring.

### A. LLM-based knowledge extraction

Structured knowledge extraction at its core relies on the ability of LLMs to comprehend, interpret, and extract meaningful information from unstructured text. By design, LLMs possess the ability to achieve strong contextual understanding, enabling them to follow user-defined instructions and produce structured outputs.

Recent research has integrated LLMs into their data extraction pipelines, often in combination with traditional natural language processing (NLP) techniques, to improve extraction quality and reduce manual intervention. This is shown by Wang *et al.*[11] where they proposed GPT-NER, a generative entity extraction framework that reformulates entity extraction as a text-generation problem. Their approach guides the LLM with special tokens and incorporates a self-verification mechanism to mitigate the effects of hallucination. Similarly, Truhn *et al.*[12] applied GPT-4 to free-text pathology reports, highlighting high corelation between GPT-4's outputs and expert-annotated data for colorectal and glioblastoma cases, suggesting that LLMs can be used as a zero-shot extractors of biomedical facts. Ateia *et al.*[13] further illustrate the power of in-context learning by developing an LLM-based system for extracting key scientific concepts, such as research questions, methods, and findings from scholarly publications. Shamsabadi *et al.*[14] and D'Souza *et al.*[15] extend these ideas to virology and ecological domains, respectively, emphasizing that LLMs can achieve competitive performance even in specialized domains, though domain adaptation still remains a challenge. Similar observation was proposed by Yanguas-Gil *et al.*[16] where they provide a valuable benchmark (ALDbench) in material sciences especially in the ALD domain. They evaluated how well LLMs can perform synthesis-related tasks in the ALD domain by asking different questions in different categories like "How to grow ZnO epitaxially on sapphire by ALD?." Their study highlights that a strong knowledge-based foundation is critical for the success in knowledge synthesis for these advanced applications. All these works show the utility of LLMs as a generative model with competitive performance on zero-shot or few-shot extraction but still requires domain adaptation.

Another aspect of structured extraction workflow is the use of a domain-specific schema or template that guides the LLM in structuring its output into machine-readable structures. In our previous work,[17] we highlight this critical role of schema-guided extraction in structuring scientific knowledge. We demonstrated how LLMs combined with domain-expert feedback can generate semantically rich process schemas that serve as blueprints for knowledge extraction. Similarly, Dagdelen *et al.*[18] trained LLMs (GPT-3 and Llama-2) together with a minimal schema, generated structured JSON outputs. Their schema-guided method enabled users to define output templates and annotate small datasets, teaching the LLM to extract and format entities and relations accordingly. This process produces highly precise, schema-compliant JSON summaries of scientific text.

These studies demonstrate the effectiveness of LLMs in extracting structured information from various data sources. However, despite their strong contextual understanding and generative capabilities, LLMs often suffer from hallucination and lack of mechanisms for validation. These limitations have motivated the emergence of agent-based workflows, which combine LLMs with different rule-based tools (such as structure validators, normalization utilities, etc.) to ensure factual accuracy, consistency, and reliability. Agentic systems extend the core functionality of LLMs by enabling them to dynamically use these tools, thereby producing more verifiable and trustworthy outputs.

### B. Agent-based workflows for knowledge extraction

Recent studies have employed agentic pipelines for different tasks, particularly for structured data extraction and validation. Li *et al.*[19] conducted a comprehensive survey of hallucination mitigation techniques and formally characterized agentic systems as architectures that integrate reasoning and action in iterative feedback loops. In their framework, hallucination is mitigated by coupling LLMs with retrieval components, for grounding responses in external evidence, planning modules, and verification agents and for self-checking or cross-checking outputs. Similarly, Sapkota *et al.*[20] discussed how multiagent pipelines, composed of specialized agents such as planners, validators, and memory modules, collectively improve accuracy by cross-verifying outputs and mitigating errors. Ghosh *et al.*[21] developed a fully agentic LLM pipeline for mining scientific literature and extracting thermoelectric and structural material properties, achieving high precision across thousands of papers. Werbrouck *et al.*[22] proposed a domain-specific agentic framework for knowledge discovery in atomic layer processing, where LLM agents interact with simulated ALD environments to hypothesize and extract latent chemical knowledge.

In this work, we utilized the contextual reasoning power of LLMs together with an agentic workflow that orchestrates multiple rule-based tools, such as schema validation, data cleaning, and chemical normalization, to extract semantically rich and verifiable knowledge from scientific publications. To our knowledge, no existing system has combined an extensive schema-based semantic validation with external knowledge normalization within an agentic framework. Our proposed workflow ensures that extracted information is not only accurate and complete but also semantically grounded and interoperable with knowledge representation frameworks such as KGs.

## III. METHODOLOGY

Our *SciKGExtract* workflow is illustrated in Fig. 2. The workflow takes as input a scientific article, a schema representation that defines the extraction objective as the set of properties and constraints for knowledge extraction from the scientific article, and optionally, a small number of domain-expert annotated examples to guide the LLM during extraction (Fig. 3). *SciKGExtract* is defined with respect to the following two steps: (1) data extraction—where the workflow uses the LLM to extract values

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```
processes:
- process_num: 1
  aldSystem.materialDeposited:
  - value: ZnO
  - supporting_sentence: The ZnO thin films were deposited ...
  processParameters.substrate:
  - value: 5x5 cm2 Si(100)
  - supporting_sentence: The ZnO thin films were deposited on 5x5 cm2 Si(100) substrates ...
  processParameters.temperature.value:
  - value: 23
  - supporting_sentence: and the substrate temperature was varied from room temperature (RT) to 140 °C.
  thicknessControl.growthPerCycle.totalCycles:
  - value: 600
  - supporting_sentence: The number of deposition cycles was fixed to 600 in all the experiments
  reactantSelection.precursor[0].precursor:
  - value: DEZ (Crompton GmbH)
  - supporting_sentence: diethylzinc (Crompton GmbH) and water were used as precursors.
- process_num: 2
  aldSystem.materialDeposited:
  - value: ZnO
  - supporting_sentence: The ZnO thin films were deposited ...
  ...
```

**FIG. 3.** Example of a domain-expert annotated extraction, illustrating how key ALD process properties and their corresponding values are represented. This structured example serves as guidance for the LLM during the extraction process.

corresponding to the properties defined by the schema directly from a scientific article and perform structure validation on the extraction against the process schema. For example, if the schema defines process temperature as a numerical property with units in °C, the validation step automatically flags extractions where the value is missing or expressed in a different incompatible units, (2) normalization—where we ground specific properties, particularly those describing chemical compounds with their corresponding PubChem Compound Identifiers (CIDs). The workflow finishes by removing null, empty, or duplicate values and produces the final normalized, schema-compliant information, which can be used for different downstream tasks. In Secs. III A–III C, we describe each of these steps in more detail.

### A. Step 1: Data extraction

The first step for the *SciKGExtract* is to generate a structured knowledge representation from a scientific article, following the predefined process schema. The step takes as input (i) a scientific article in the text or markdown format and (ii) a process schema that serves as a structure model for structuring the extracted information. The process schema can be either manually designed by domain-experts, leveraging their expertise and experience with the domain, or derived from an automated framework such as $Schema - Miner^{pro}$,[23] which integrates LLMs with domain-expert validation to curate semantically rich process representations. Optionally, the step can also incorporate a third input, (iii) domain-expert-annotated examples (Fig. 3) that serve as gold-standard extractions, providing few-shot guidance to the LLM that has been shown to work effectively in practice. In few-shot or in-context learning,[24] a small number of example input, output pairs are embedded directly into the LLM's prompt. The LLM then uses these examples to infer the desired extraction pattern and output structure without the need for retraining the LLM with the desired outputs.

These inputs are then passed to the LLM together with an instructional prompt (see the full prompt in Sec. 1 in the supplementary material) that defines the extraction task. The prompt is designed to guide the model through three main components:

1. **Context specification**: Here, we specify the process definitions, task description showing the extraction scope.
2. **Extraction instructions**: Here, we describe the reasoning steps for the model, for example, identifying relevant sentences, interpreting abbreviations, normalizing numerical values (e.g., temperatures in °C), and preserving units.
3. **Input and output formats**: Finally, we specify the input formats, for example, text representation of the scientific article, and a process schema to enforce a deterministic structure output.

In this paper, we adopted JSON as the target representation format for representing structured knowledge due to its human-readable and machine-readable structure, as well as its support for schema validation. The extracted structured knowledge is validated against the provided process schema that is necessary due to the nondeterministic nature of LLMs and their potential to occasionally omit or misinterpret instructions. This step employs a rule-

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based schema-validation package, jsonschema in PYTHON, to ensure compliance with the schema. In essence, the validator verifies that all required properties have been extracted and that their corresponding values adhere to all the constraints. If the extracted information fails to conform to these conditions, the step is repeated with the feedback to the LLM for correction and reprocessing; otherwise, it proceeds to the next step for normalization.

## B. Step 2: Normalization

Normalization is the process of standardizing and resolving extracted data entities with an authoritative data source so that identical or equivalent entities are represented consistently across the dataset. In the context of scientific publications, the need for normalization arises due to inconsistent textual representation of the same chemical compound. For example, *water* and $H_2O$ refer to the same compound and should be normalized to a single canonical representation. Similarly, another example includes the ALD precursor *diethylzinc*, which may appear as *diethyl zinc (DEZ)*, *DEZn*, or *ZnEt2* across different publications.

In this study, as we focus on ZnO and *IGZO* ALD processes, the extracted information includes several properties that contain chemical compounds and molecular formulas, such as precursors, coreactants, and carrier gases. We, therefore, used the PubChem repository as an authoritative source for normalization due to its comprehensive coverage of different chemical compounds and supports an open-access application programming interface (API) end point for effective programmatic querying. To enable consistent referencing and downstream analysis, each of the textual variations discussed before should be mapped to a single, persistent identifier. For instance, all representations of *diethylzinc* should be linked to the PubChem identifier (CID) of 101667988 and water with 962. This consolidates all the associated metadata and allows equivalent compounds to be grouped, compared, and queried uniformly across different studies.

In *SciKGExtract*, the normalization procedure (as shown in Fig. 2) is performed stepwise based on the complexity and the nature of the extracted chemical entity. The steps include the following:

### 1. PubChem API search

We first start by querying the extracted compound with the PubChem API. We used two different end points provided by PubChem based on different search operations: SEARCH BY COMPOUND NAME and SEARCH BY MOLECULAR FORMULA. If a match is found from either of these end points, PubChem returns the corresponding CID and a list of synonyms associated with the requested chemical compound. Internally, the PubChem maintains a list of synonyms for each chemical entity, which the API uses to resolve different textual variants with the same CID, enabling the unification of equivalent chemical entities. But still with this advance searching and querying capabilities provided by PubChem, some of the chemical representations fail to be linked with a unique CID value and require an extensive heuristic-based search.

### 2. Heuristic-based search

In the case where the PubChem API fails to find a valid CID, an additional heuristic-based search is performed. The heuristics include two different components, and based on the complexity of chemical compound, one or both of the components can be used to find a valid CID for the requested compound.

### 3. PubChem local lookup

PubChem as a service provides a local, downloadable copy of its database: CID-Synonym-filtered. This file provides a mapping between different CID values and their corresponding chemical synonym values. As part of this first heuristics, we apply more advanced searching operations (e.g., LMDB Indexed Search, Fuzzy matching via rapidFuzz) between the query compound and all the chemical synonyms present in the local PubChem database to retrieve candidate matches. These different searching techniques utilize different similarity metrics like Levenshtein distance[25] that measures the difference between two chemical entities. Based on a preferred threshold value (e.g., similarity ¿= 85%), we can narrow down to a small subset of candidate entities in the database and consequently to a valid CID.

### 4. LLM-assisted disambiguation

The second component as part of heuristic-based search is initiated when no matching CID is found by PubChem Local Lookup. The workflow triggers a disambiguation loop for the requested chemical compound where a secondary LLM is used to propose an alternative or a more formal representation of the chemical compound. For example, the extracted substrate: *Thermally grown SiO2-buffered substrate* contains an explicit chemical compound, $SiO_2$, which can be independently resolved against PubChem. The LLM identifies and returns $SiO_2$ as the relevant chemical compound and loops back to the normalization procedure (see Fig. 2—Normalization) to retrieve the corresponding CID: 24261. The final normalized representation retains the original extracted value: *Thermally grown SiO2-buffered substrate*, while associating it with the resolved CID 24261, thereby preserving both semantic context and enabling consistent chemical grounding.

To achieve this formal representation, a secondary LLM (e.g., gpt-5) is prompted with the extracted value and with the whole extracted information as a context to propose an alternate representation. Contextual information helps the LLM resolve ambiguities and propose a valid chemical representation. The revised representation is resubmitted to the PubChem API for another normalization attempt. If no valid match is found after this iteration, the compound is marked as unnormalized.

Finally, after the normalization procedure is finished and a valid PubChem CID is found, a sameAs property has been added for the chemical compound containing the PubChem URI (as shown in Fig. 4).

## C. Cleaning

The final step of the *SciKGExtract* workflow is to clean the normalized, schema-compliant information by removing empty and duplicate values. This step is necessary as LLMs for certain

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```
"reactantSelection": {
    "precursor": [
        {
            "compound": {
                "value": "ZnO",
                "sameAs": ["https://pubchem.ncbi.nlm.nih.gov/compound/14806"]
            },
            "precursor": {
                "value": "diethylzinc (DEZ)",
                "sameAs": ["https://pubchem.ncbi.nlm.nih.gov/compound/101667988"]
            }
        },
        {
            "compound": {
                "value": "Al2O3",
                "sameAs": ["https://pubchem.ncbi.nlm.nih.gov/compound/9989226"],
            }
            "precursor": {
                "value": "trimethylaluminum",
                "sameAs": ["https://pubchem.ncbi.nlm.nih.gov/substance/241178067"]
            }
        }
    ],

    "carrierGas": {
        "value": "Ar",
        "sameAs": ["https://pubchem.ncbi.nlm.nih.gov/compound/23968"]
    }
    "purgingGas": {
        "value": "Argon",
        "sameAs": ["https://pubchem.ncbi.nlm.nih.gov/compound/23968"],
    }
}
```


**FIG. 4.** Example of an extracted and normalized ALD process information represented using the JSON format. The purpose is to show how different chemical entities have been normalized with PubChem CIDs to group similar entities together.

properties can generate incomplete or empty values after extraction that do not align with the target domain. Additionally, since not all scientific publications report all of their process properties in their text, it can become difficult for the LLM to extract values for these properties. As a result, they output empty placeholders to denote this behavior. For example, *{"growth_rate": "NOT FOUND"}* shows that no growth per cycle (*GPC*) value was found in the scientific article. As a result, it becomes necessary to remove all empty values from the extracted information to ensure overall data quality and consistent information. The cleaning process is applied as a rule-based procedure, implemented as a Python module, which systematically validates and filters the final structured information, thereby improving its suitability for different downstream tasks such as visualization and knowledge graph integration.

## IV. EXPERIMENTS AND RESULTS

It is typical to AI system workflows to conduct experimental evaluations to demonstrate their performance. In this context, the term "experiment" refers to assessing the proposed approach, rather than performing physical ALD experiments. Therefore, in this section, we present the evaluation of the proposed *SciKGExtract*, focusing on its ability to extract structured knowledge from real-world ALD experimental processes, reported in different scientific papers. Sections IV A–IV C describe the dataset and experimental setup (including the LLMs and development frameworks). We further present analyses of the extracted structured data, assessing the accuracy, consistency, and semantic correctness of the outputs. Finally, we analyze the extracted ALD data to illustrate the types of insights that become accessible through structured representations, highlight the opportunities enabled by systematic normalization of ALD process information, and clarify current limitations of LLM-based extraction.

### A. Dataset

The AtomicLimits ALD Database[9] is an online, community-driven repository that compiles literature-reported ALD processes.

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Launched by Kessels *et al.* in 2019, it provides detailed, manually curated information on ALD-deposited materials, precursors, and coreactants, all of which are submitted by contributors from the ALD community and subsequently reviewed by moderators to ensure consistency and data quality. In parallel, the AtomicLimits ALE Database[26] documents atomic layer etching (ALE) processes reported in literature, including the etched material, precursor, and coreactant used, and whether the process is isotropic or anisotropic. Due to its manual curation and verified data quality, the AtomicLimits ALD Database provides a gold-standard source of ALD knowledge for assessing the performance of our workflow.

To implement this, we used a representative subset of ZnO and *IGZO* ALD publications obtained from the AtomicLimits ALD Database. We selected ZnO as one of the test cases because it is among the most extensively studied materials in ALD,[27] with the AtomicLimits ALD Database listing, at the time of writing, 30 distinct ALD chemistries (precursor/coreactant pairs) for depositing ZnO. As a wide-bandgap semiconductor with high transparency and controllable electrical properties, ZnO is an attractive material for ALD research, with applications including thin-film transistors, solar cells, and sensors.[28,29]

As a second case study, we also considered *IGZO*, which is a quaternary oxide semiconductor that has attracted significant attention in recent years due to its high carrier mobility, compatibility with low-temperature processing, and high optical transparency.[30] One of the approaches to deposit *IGZO* using ALD is the supercycle method, which includes alternating the three binary ALD processes of $In_2O_3$, $Ga_2O_3$, and ZnO in a supercycle.[31] This method allows precise tuning of film composition and resulting properties by adjusting the number of subcyles and their sequence.[32] However, the design space for *IGZO* supercycle processes is extremely large due to the choice of binary cycle sequence, cycle ratio, precursors, coreactants, and process conditions. Therefore, the space of possible process combinations expands rapidly as additional parameters and supercycle design choices are introduced.

Quaternary and complex ALD systems offer significant opportunities for optimization, yet this task makes manual process design increasingly difficult. This highlights the need for structured, machine-readable datasets that include not only binary ALD processes but also more complex supercycle systems. With such datasets, data-driven approaches—potentially including machine learning—could be used to navigate and optimize the high-dimensional parameter space of multicomponent ALD processes, enabling more efficient exploration of new chemistries, growth mechanisms, and finally the discovery of novel materials. Therefore, including IGZO as a case study demonstrates both the capabilities of our extraction workflow and the broader potential of structured ALD data to support future data-driven design of complex materials.

After obtaining the ZnO and *IGZO* ALD papers from the AtomicLimits database in PDF format, preprocessing is necessary to prepare them for structured data extraction. Each scientific article was processed using an automated PDF-to-text conversion pipeline to prepare it for structured information extraction. Specifically, we used Docling,[33] for parsing text and tables, and the extracted text was saved in a markdown format to preserve the table structures in the text. This processed corpus was then passed to the *SciKGExtract* workflow for schema-guided knowledge extraction.

## B. Experimental setup

### 1. Agentic workflow framework

The proposed *SciKGExtract* workflow have been implemented using the LangGraph framework, which has been designed for building and orchestrating agent-based AI systems. LangGraph offers a modular architecture for defining agents and tools, utilizing LangChain at its core to interface with both open-source and closed-source LLMs.

### 2. LLMs tested

For this study, we used OpenAI's GPT-4o and GPT-5-mini models as the primary LLMs for structured knowledge extraction from scientific articles. These state-of-the-art models support agentic workflows and can natively produce outputs in structured formats such as JSON. Since the OpenAI models are accessed through API end points, no dedicated GPU or specialized hardware infrastructure was required for experimentation. JSON was chosen as the output format for representing structured knowledge due to its simplicity, hierarchical representation, and interoperability with downstream tasks. Finally, on average, the processing cost per paper was approximately 0.20 USD, considering all workflow components including extraction and normalization.

### 3. ALD experimental schema

As discussed in the beginning of Sec. IV, extracting structured information from ALD literature requires a schema representation that defines the structure, properties, and constraints of the process to be captured across different publications. For this purpose, we employed the experimental ALD Schema,[34] which was developed using our previous work.[23] This schema was generated using an agentic workflow that combines iterative LLM-based schema extraction and ontology grounding with domain-expert validation for a complete and semantically rich representation of the target process.

### 4. PubChem normalization module

To ensure normalization of extracted chemical entities, we integrated our workflow with PubChem that provides multiple access mechanisms, including API calls and bulk data downloads. In our implementation, we utilized both the PubChem REST API and a CID-to-synonym mapping file to normalize extracted chemical names and map them to their corresponding CIDs. Furthermore, the workflow has been designed to be modular and extensible, allowing the incorporation of other external data repositories beyond PubChem for different target domains.

### 5. Evaluation metrics

To assess the performance of the *SciKGExtract* workflow, we used two commonly adopted evaluation metrics used

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extensively in literature for comparing pairwise similarity between different extractions:

- **BERTScore (BERT F1)**[35] is an embedding-based metric that measures the semantic similarity between the extracted and the reference annotations. It uses pretrained models to compare the contextual representation of the predicted and the reference text. This metric is important for scientific literature as it can compare more precisely the same chemical compounds described using different names or abbreviations.
- **Accuracy** measures how often the extracted value exactly matches the reference annotations. An extracted value is considered correct, if it is identical to the reference annotation. Accuracy in semantics is considered a strict measure of correctness and is particularly useful for evaluating the normalization task, where precise CIDs of chemical compounds have to be matched.

## C. Results and discussion

In this section, we present the qualitative and quantitative results of applying the *SciKGExtract* workflow on our ALD literature dataset. First, we provide a quantitative comparison between the *SciKGExtract* extractions and the gold-standard domain-expert annotated extractions to evaluate the accuracy and reliability of our system. Finally, we present a qualitative analysis by providing a brief exploratory analysis to highlight trends, patterns, and insights observed in the ZnO and *IGZO* ALD literature, demonstrating a practical utility of structured, schema-guided knowledge extraction.

### 1. Quantitative evaluations

For quantitative evaluations, we used three experimental settings, where we first compared the values extracted by *SciKGExtract* against the existing annotations (material deposited, precurors, and coreactants) available in the AtomicLimits Database. In the second experimental setting, we compare the normalized PubChem CIDs by *SciKGExtract* with manually assigned PubChem CIDs by the domain-experts. Finally, in the third experimental setting, we compare for two scientific articles, the extraction done by *SciKGExtract* and by the domain-experts. These comparisons enabled us to evaluate both surface-level similarity and semantic coherence of the extracted information. **Experimental Setting 1: Comparing *SciKGExtract* extracted values with AtomicLimits annotations.** The AtomicLimits Database, from which our dataset was curated, provides limited annotations for each publication, specifically materials, precursors, and coreactants used in the reported ALD process. This makes it ideal for assessing the extraction quality of our *SciKGExtract* workflow.

Table I summarizes the results of our first comparison, showing scores for two LLMs: GPT-4o and GPT-5-mini across both ZnO and *IGZO* ALD processes. The results indicate moderate to high coherence and alignment of the extracted values with the AtomicLimits annotations. The materials extraction in ZnO ALD processes shows highest coherence with the AtomicLimits database annotations, achieving BERT F1 score of **95.53** by GPT-4o and **96.32** by GPT-5-mini. Similar coherence was also observed for the coreactant extractions in *IGZO* ALD processes, achieving BERT F1 score of **73.13** by gpt-4o and **86.95** by gpt-5-mini. Comparing across models, GPT-5-mini consistently outperforms GPT-4o for both ZnO and *IGZO* datasets and across all properties, suggesting that GPT-5-mini captures domain-specific context and semantic relationships more effectively.

**TABLE I.** Quantitative evaluation of extracted properties of ZnO and *IGZO* ALD processes compared with AtomicLimits Database annotations using BERT-F1 metrics (expressed in percentages %). Bold values indicate the highest BERT F1 achieved between two LLM models.

| | ZnO | | IGZO | |
|---|---|---|---|---|
| | GPT-4o | GPT-5-mini | GPT-4o | GPT-5-mini |
| Material | 95.53 | **96.32** | 77.71 | **79.03** |
| Precursor | **66.52** | 65.51 | 58.51 | **58.61** |
| Coreactant | 76.30 | **83.33** | 73.13 | **86.95** |

**Experimental Setting 2: Evaluating *SciKGExtract* Normalization Accuracy**. To evaluate the effectiveness of the normalization step in the *SciKGExtract* workflow, we assessed the accuracy of PubChem CID assignments for the extracted chemical entities. Table II presents a comparison between the normalized CIDs produced by our workflow and the domain-experts annotated CIDs of the AtomicLimits Database. The evaluation was performed using the *Accuracy* metric, which measures the number of correctly matched CIDs relative to the total number of annotated entities.

Overall, the results indicate that the normalization step substantially enhances the correctness and consistency of extracted chemical representations. For both ZnO and *IGZO* processes, the workflow performed better in normalization of properties *material*, *precursor*, and *co-reactant* properties and demonstrated high accuracy, with values exceeding 80% for both GPT-4o and GPT-5-mini. Between the two models, GPT-5-mini achieved marginally higher accuracy for the *precursor* property (**82.70** for ZnO and **100** for IGZO). At the same time, GPT-4o performed slightly better for the *coreactant* property in ZnO processes. These findings highlight the impact of CID-based normalization in unifying chemically equivalent expressions and ensuring consistent entity resolution. **Experimental Setting 3: Comparing extracted values with gold-standard domain-expert annotations**. We annotated with the help of a domain-expert, two scientific articles, one for ZnO, titled as "Low-temperature atomic layer deposition of ZnO

**TABLE II.** Comparison of normalized PubChem CIDs for ZnO and *IGZO* ALD processes compared with manually annotated PubChem CIDs for AtomicLimits Database annotations using accuracy score. Bold values indicate improved scores after normalization.

| | ZnO | | IGZO | |
|---|---|---|---|---|
| | GPT-4o | GPT-5-mini | GPT-4o | GPT-5-mini |
| Material | **100** | **100** | **100** | **100** |
| Precursor | 81.61 | **82.70** | 96.80 | **100** |
| Coreactant | **81.85** | 75.68 | 89.81 | **93.93** |

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**TABLE III.** Comparison of structured extractions of two scientific articles (one each for ZnO and *IGZO*) between *SciKGExtract* and domain-expert annotated gold-standard extractions. The comparison includes the number of processes extracted and the semantic similarity (BERT F1) between the extractions. The bold BERT F1 score shows the higher value between GPT-4o and GPT-5-mini.

| | ZnO | | | IGZO | | |
|---|---|---|---|---|---|---|
| | Number of Processes | | | Number of Processes | | |
| | **Gold-Standard** | **SciKGExtract** | BERT F1 | Gold-Standard | SciKGExtract | BERT F1 |
| GPT-4o | 8 | 5 | 78.80 | 5 | 1 | **87.35** |
| GPT-5-mini | 8 | 2 | **86.44** | 5 | 5 | 85.01 |

thin films: Control of crystallinity and orientation,"[36] and the other for *IGZO*, titled as "Amorphous IGZO TFT with High Mobility of 70 $cm^2$/Vs via Vertical Dimension Control using PEALD."[37] These two scientific articles provide us an opportunity to compare and evaluate the quality of extractions performed by our *SciKGExtract* workflow with the domain-expert assisted extractions.

Table III summarizes the results for this comparison. We first evaluate the total number of distinct processes reported in the scientific article. An ALD scientific article can report multiple ALD processes as part of their analysis where different processes can differ by a single setting such as deposition temperature or by a single observation such as GPC value. Structured extraction supports this behavior by outputting a list of ALD process instances where an instance can differentiate from the other by a single or multiple values. The results shows that eight different processes were extracted for the ZnO paper and five for the *IGZO* one by the domain-expert that serve as the gold-standard annotations. GPT-5-mini performed comparatively better as compared to GPT-4o in extracting all different processes from the scientific article.

The other evaluation we performed was to evaluate the semantic similarity between the extraction by domain-expert and the *SciKGExtract* using the BERT F1 score. GPT-5-mini, on average, performed better in extracting the correct values for different properties and show high correlation with the processes extracted by the domain-experts.

However, we observe that several processes are still missed by the workflow. This is primarily due to (i) implicit reporting of process variations in the text, where changes in parameters such as temperature or GPC are described narratively rather than as clearly separated processes, and (ii) inconsistencies in how experimental conditions are presented across papers (e.g., distribution across text, tables, and figures). These factors highlight the limitations of single-pass extraction and motivate the need for iterative evaluation and refinement mechanisms, which we identify as an important direction for future work.

### 2. Qualitative evaluations

For qualitative evaluations, we will first explore the *SciKGExtract* extraction errors that have been identified by the domain-experts and some of the possible reasons behind them. Second, we will discuss briefly about the relative frequencies of the properties extracted by the framework, highlighting which properties were mentioned the most and least in an ALD scientific publication. Third, we will discuss and analyze in detail some of the important insights from the ZnO and *IGZO* ALD process extraction data using our *SciKGExtract* workflow.

*a. SciKGExtract extraction errors recorded for ZnO and IGZO processes.* Table IV presents a summary of key ALD parameter values extracted from scientific literature of ZnO ALD processes. The table lists the most frequently occurring values for categorical properties, such as precursor and coreactant. For numerical properties, including deposition temperature and GPC, the minimum and maximum values across the dataset are reported. This summary allows us to rapidly assess the extracted parameter space for ZnO ALD and, thereby, detect dominant precursors and coreactants, typical process temperature ranges, and potential extraction anomalies before performing more detailed analysis.

A closer look at the extracted precursor and coreactant names for ZnO and *IGZO* ALD processes reveals several entries that do not belong to these specific chemistries. For instance, trimethylaluminum (*TMA*) and trimethylgallium (*TMGa*)—precursors typically used for depositing aluminum oxide ($Al_2O_3$) and $Ga_2O_3$, respectively—and hydrogen sulfide ($H_2S$), which is not an oxygen source as a coreactant, appear in the extracted dataset. These arise because the model occasionally captures reactants for ZnO mentioned elsewhere in the same publication, such as those used for doped films, multicomponent stacks, or supercycle sequences, rather than restricting itself to the ZnO-specific ALD processes. This underscores the importance of prompt constraints that explicitly limit extraction to ZnO ALD properties, preventing the inclusion of parameters from other ALD processes reported in the experimental workflow. Although data validation and refinement is not part of this work, they will be incorporated in future work to address this issue and improve the overall data quality.

Regarding the ZnO ALD deposition temperature, the extracted dataset spans from 23 to 600 °C. In the literature, ZnO ALD deposition temperatures have been reported over a broad range (approximately 25–500 °C), depending on the precursor chemistry employed. However, these reported extremes should be interpreted with caution, as the temperature window associated with true ALD behavior—defined by self-limiting surface reactions—is substantially narrower. For the widely used diethylzinc (*DEZn*)/water (*H2O*) process, the ALD window has been reported in the range of approximately 110–170 °C.[28] At elevated temperatures, precursor decomposition and gas-phase reactions can

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**TABLE IV.** Overview of extracted ZnO and *IGZO* ALD process properties obtained using the SciKG_Extract workflow with the GPT-5-mini model. The values in RED shows the extraction errors in ZnO and IGZO processes as identified by the domain-experts.

| | ZnO | IGZO |
|---|---|---|
| Precursor | *DEZn*, *DMZn*, *TMA*, *TMGa* | *DADI*, *DEZn*, *TMA*, *TMGa*, *TEIn* |
| Coreactant | $CO_2$, *EtOH*, $H_2O$, $H_2S$, $NH_4$, $N_2O$, $O_2$ *plasma*, $O_3$ | $O_2$ *plasma*, $O_3$, $O_2$/*Ar plasma*, $H_2O$ |
| Deposition Temperature (°C) | 23–600 | 100–300 |
| GPC (Å) | 0.06–115 | *In*: 0.08–2.1, *Ga*: 0.18–1.97, *Zn*: 0.12–1.28 |

contribute to film growth, leading to mixed ALD–CVD behavior rather than purely self-limiting ALD.[38]

The extracted GPC values for ZnO ALD exhibit an unusually broad range (0.06–115 Å), which is orders of magnitude larger than expected for ALD. In ALD, GPC refers to the film thickness increase per cycle, which typically ranges from 0.5 to 1.5 Å, depending on the process used.[8] For ZnO ALD, literature-reported GPC values generally fall between approximately 0.5–3 Å depending on the precursor chemistry, with the widely used *DEZn*/*H2O* process most commonly yielding 1.8–2.0 Å within the ALD window.[28,39] The presence of extremely large extracted GPC values is thus attributed to an extraction or reporting error, where properties mentioned in the specific paper were incorrectly interpreted as GPC. A closer inspection revealed that the GPC value of 115 Å originates from a statement reporting mass uptake per cycle (115 ng/cm$^2$/cycle), which was incorrectly interpreted as a thickness-based GPC value. This highlights a specific failure mode where the workflow confuses mass-based growth rates with thickness growth per cycle due to similar contextual phrasing. In contrast, the extracted GPC values for the individual *IGZO* binary oxides ($In_2O_3$, $Ga_2O_3$, and ZnO) fall within physically reasonable ALD GPC ranges.[39]

Figure 4 illustrates a representative JSON snippet from the extracted output of a specific ZnO ALD publication, showing how the *SciKGExtract* structures values according to the defined schema. This particular example also demonstrates that the workflow captures ALD information for doped or mixed ZnO films—in this case, Al-doped ZnO—when such processes are described in the same paper. While this reflects the model's ability to detect multiple ALD chemistries, it also highlights the need for stricter extraction constraints to ensure that only ZnO-specific process parameters are included when building a ZnO-focused dataset. A more detailed quantitative and comparative analysis of these extracted properties is provided in Secs. IV C 2 b and IV C 2 c.

*b. Relative frequency of ALD schema properties in ZnO and IGZO* processes. We analyze the relative frequency of different schema properties to assess their coverage across the extracted processes. For both GPT-4o and GPT-5-mini, we observe a consistent trend in the most frequently extracted properties, namely, material deposited, carrier and purging gases, substrate, ALD method, precursor, and coreactant. These properties are commonly reported in ALD literature and are reliably identified by the extraction workflow.

In contrast, properties such as growth-per-cycle (GPC) values, dosing time, and purging time are among the least frequently extracted. This reflects their lower reporting frequency or variability in how they are described in scientific publications. While both models exhibit similar trends in property coverage, GPT-5-mini extracts a larger number of processes overall compared to GPT-4o, resulting in a higher absolute count of extracted properties.

*c. Insights from extracted ZnO and IGZO* ALD data. Apart from quantitative evaluations shown in Secs. IV C 2 a and IV C 2 b, it is important to demonstrate the practical value of structured data extraction from the ALD literature. This conversion of heterogeneous ALD publications into a unified structured format enables rapid exploration of trends, outliers, and variability across studies, which would be impractical to identify manually. In a future use case, such extracted datasets can support ALD process development by providing insights, for a given process, into commonly employed chemistries, expected GPC values and typical deposition temperature windows across literature.

To illustrate these opportunities that arise for the ALD community, we next analyze in more detail the ZnO and *IGZO* ALD datasets generated by our *SciKGExtract* workflow, focusing specifically on three core ALD parameters: (1) ALD method, (2) precursor and coreactant selection, and (3) GPC as a function of deposition temperature. Analysis of GPC trends is restricted to ZnO, where the larger and more mature literature yields sufficient extracted data to enable meaningful cross-study comparison.

Before presenting these results, it is important to clarify both the data filtering strategy and the interpretation of the extracted data, as the analysis depends strongly on the structure of the underlying extraction schema.[23] In this work, the term *"study"* refers to a distinct ALD process extracted from the literature, rather than to an individual publication. Consequently, a single paper may contribute multiple studies when several ALD methods, precursor chemistries, or process conditions are reported.

Additionally, this analysis focuses exclusively on literature reported ALD processes of binary ZnO and quaternary *IGZO* films. ALD processes involving doped ZnO or materials other than ZnO and IGZO identified during data extraction were excluded to ensure consistency across the dataset. That was performed by filtering the extracted dataset, using *"materialDeposited"* property, including only entries labeled as ZnO or *IGZO*, respectively. Additionally, when analyzing the precursor and coreactant choice in *IGZO* studies, the data were filtered at the suboxide level to reflect the individual oxides comprising the *IGZO* supercycle, namely, $In_2O_3$, $Ga_2O_3$, and ZnO. This was achieved by filtering the *"compound"* property according to each oxide (e.g., $In_2O_3$, $Ga_2O_3$, or ZnO)

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Regarding the trends of ALD methods used in ZnO and *IGZO* ALD publications, the results are presented in Fig. 5. For ZnO [Fig. 5(a)], thermal ALD overwhelmingly dominates, consistent with the maturity and reproducibility of the well-established *DEZn*/$H_2O$ chemistry, with understood self-limiting reactions and reproducible growth behavior.[40] Plasma-enhanced ALD (PEALD) remains less frequently reported than thermal ALD in the ZnO literature, despite the potential advantages of PEALD for ZnO, including enhanced reactivity and lower deposition temperature processing.[41] In addition to thermal ALD and PEALD, a limited number of studies employing spatial ALD (SALD) is present in the extracted dataset. The comparatively low representation of SALD reflects its primary use in high-throughput and industrial-scale deposition, whereas the majority of the analyzed literature focuses on laboratory-scale thin-film deposition.[42]

On the other hand, *IGZO* [Fig. 5(b)] shows a much smaller dataset and a stronger reliance on PEALD than thermal ALD. This likely reflects the challenges associated with depositing multicomponent oxide films using thermal ALD, due to unexpected chemical reactions, complicating the reliable control over film composition and thickness.[43] In contrast, PEALD provides increased surface reactivity, enabling higher growth rates at lower temperatures while suppressing interlayer diffusion relative to conventional thermal ALD.[44]

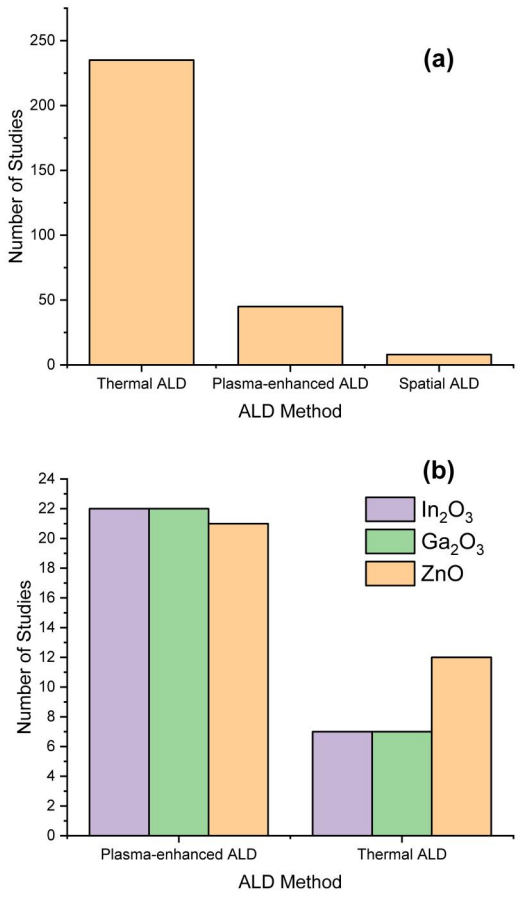



**FIG. 5.** Distribution of ALD methods reported in literature for the deposition of (a) ZnO and (b) *IGZO* films, expressed as the number of studies.

Beyond the choice of the deposition method, reactant selection plays a central role in determining film quality, composition, and growth characteristics. As shown in Fig. 6, structured extraction enables the visualization of precursor and coreactant choice across the literature, revealing trends that can inform early-stage process development. For ZnO [Figs. 6(a) and 6(b)], the strong dominance of *DEZn* with $H_2O$ reflects a highly mature ALD chemistry. In contrast, the broader distributions observed for *IGZO* ALD [Figs. 6(c) and 6(d)] highlight the increased complexity of multicomponent oxide deposition, where multiple metal-organic precursors are explored. For the coreactants, the data indicate a strong reliance on $O_2$ plasma, followed by ozone ($O_3$). Suboxide analysis further reveals that while zinc chemistry in *IGZO* largely mirrors binary ZnO, $Ga_2O_3$ deposition relies on a limited number of precursors, whereas $In_2O_3$ exhibits a broader range of precursor chemistries.

The resulting ZnO and *IGZO* ALD datasets also allow the assessment of the quality of the *SciKGExtract* workflow. As already discussed before, a small number of chemically inconsistent precursors and coreactants were identified, including cases where $H_2O$ or *TMA* were incorrectly extracted as zinc precursors [Fig. 6(a)] or where species such as $NH_3$ appeared among coreactants despite not serving as oxygen sources for ZnO ALD [Fig. 6(b)]. These instances arise both from ambiguities in the source ALD publications—where chemical species are discussed in a broader experimental context rather than being clearly assigned specific ALD roles—and from limitations of the automated extraction workflow, which can misattribute chemical species. Importantly, such cases were infrequent and did not affect the dominant trends observed. Their identification underscores the importance of systematic validation of extracted ALD datasets and highlights the need for clearer ALD-specific terminology in the literature, as well as continued refinement of extraction instructions and schemas.

In addition to extraction quality, the effectiveness of the normalization approach must also be assessed. Normalization successfully reduced chemical diversity arising from naming variations in ALD publications, as illustrated by widely used precursors such as *DEZn*, *TMGa*, and trimethylindium (*TMIn*) in Fig. 6. However, some limitations of automated normalization were observed when applied to ALD-specific chemistry. More specifically, chemically similar but nonidentical precursors—such as *TMIn* and triethylindium (*TEIn*) or *TMGa* and triethylgallium (*TEGa*)—were occasionally mapped to the same CID due to incomplete or ambiguous precursor naming in the source literature, which triggered synonym-based normalization despite underlying chemical differences [Fig. 6(c), Table S1].

In addition, coreactants such as mixed-gas plasmas (e.g., $O_2$/*Ar* plasma) and ill-defined species (e.g., synthetic air) could not be assigned a CID (Table S2). In certain cases, $O_2$ plasma was extracted as $O_2$ and, therefore, mapped to the same CID as molecular $O_2$; for the purposes of the present analysis, these entries were

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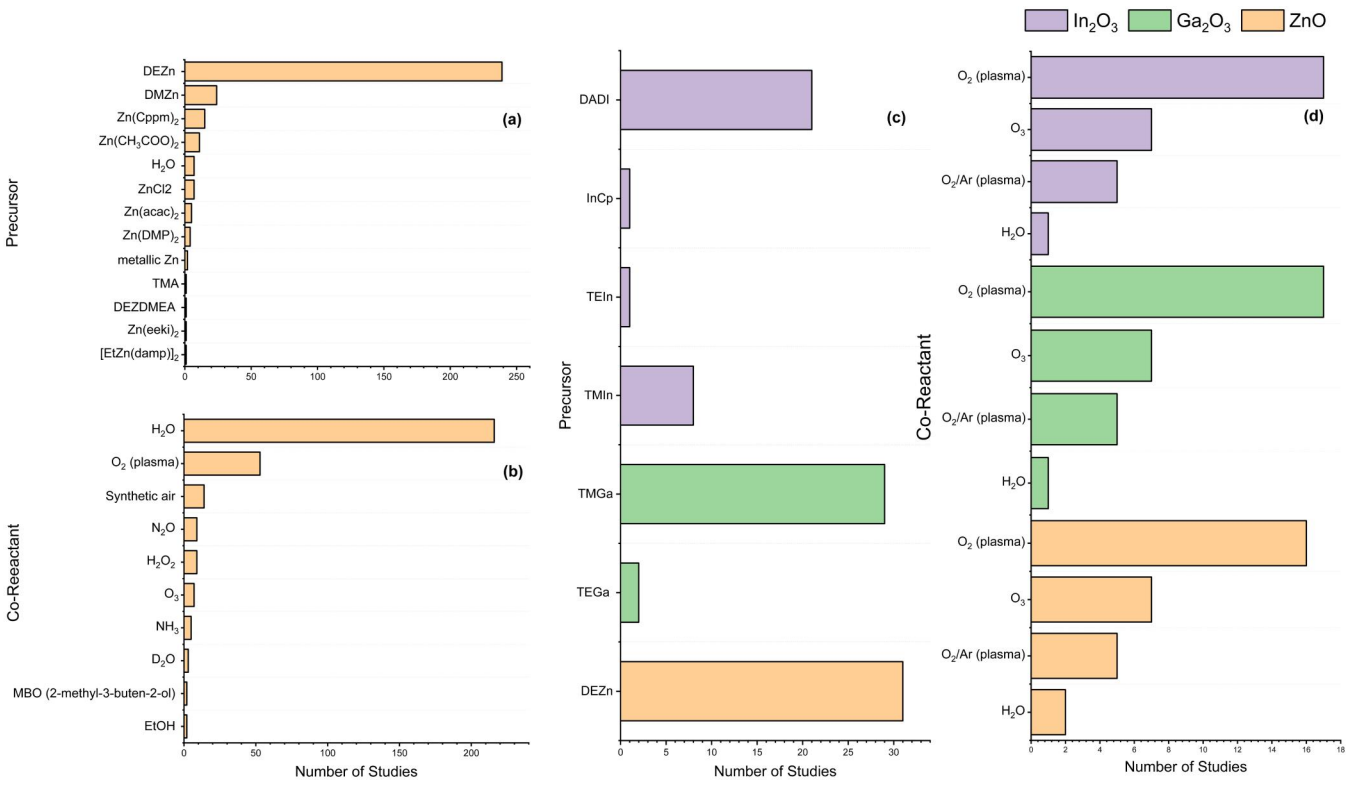



**FIG. 6.** Distribution of precursors and coreactants used in ZnO and *IGZO* literature shown as number of studies. (a) and (b) summarize the zinc precursors and coreactants used for ZnO ALD, respectively, highlighting the strong dominance of *DEZn* with $H_2O$ as the primary coreactant. (c) and (d) show the corresponding precursor and coreactant distributions for *IGZO* ALD and more specifically at the suboxide level for $In_2O_3$, $Ga_2O_3$, and ZnO. *IGZO* data reveal a greater diversity of chemistries compared to ZnO, relying on $O_2$ plasma as the coreactant. Chemical species were normalized during the extraction workflow, with the corresponding PubChem CIDs provided in the supplementary material.

merged. While these cases remain infrequent, they underscore the need for domain-aware normalization rules and systematic evaluation to ensure accurate representation of chemical diversity in large-scale ALD data analysis.

Figure 7 summarizes the extracted GPC of ZnO *DEZn*/$H_2O$ ALD process, as a function of deposition temperature across the ALD literature. The majority of reported data cluster between approximately 100 and 250 °C, with GPC values centered around 1–2 Å. The extracted GPC values are consistent with trends in literature, which commonly estimate GPC values of approximately 1.8–2.0 Å for thermal ZnO ALD using *DEZn* and $H_2O$.[28,45] This agreement indicates that the automated extraction workflow preserves the expected growth behavior reported across the ALD literature.

Regarding the temperature ranges defining the ALD window, literature values vary substantially across studies, with ranges such as 70–200, 125–150, and 150–200 °C being reported.[28,46] This variability is clearly reflected in Fig. 7 and highlights inconsistencies in how ALD windows are defined and reported in literature. Such differences may arise from variations in results interpretation, as well as differences in thickness characterization techniques, reactor configuration, sample positioning, and temperature measurement, all of which must be considered when comparing studies.

In addition to this broad clustering, Fig. 7 also reveals outliers in both deposition temperature and GPC, which have been discussed before. For example, deposition temperatures exceeding 400 °C and GPC values approaching 4.5 Å are observed. These points may reflect ambiguities during extraction, such as the inclusion of postdeposition annealing temperatures, misreported or nonsaturated growth rates, or extraction errors. Such cases emphasize the importance of well-defined ALD terminology and the application of strict instructions during automated data extraction.

Despite these limitations, the aggregated dataset provides a robust overview of thermal ZnO ALD growth behavior by capturing both the dominant deposition regimes and expected deviations across studies. Importantly, this analysis also highlights key requirements for improving the usability of literature-derived ALD data. While the present dataset enables rapid identification of typical growth regimes and outliers, more detailed metadata—such as the methods used for film thickness characterization, reactor configuration, dose and purge timings, and explicit verification of saturation behavior—would be required to support more

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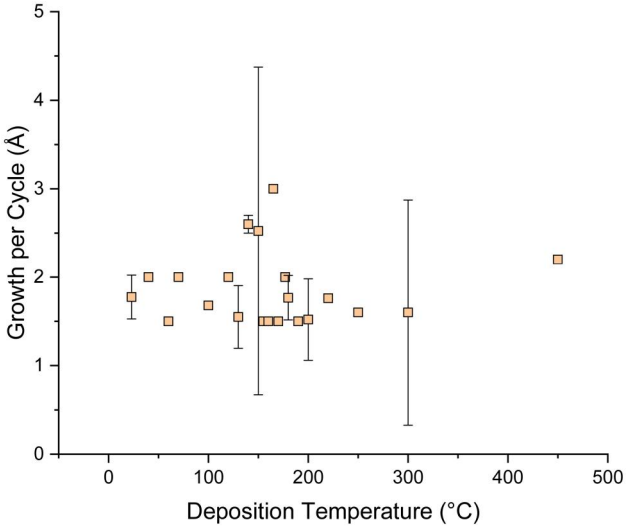



**FIG. 7.** Extracted GPC of ZnO deposited by *DEZn*/$H_2O$ ALD process as a function of deposition temperature across the literature. For temperatures at which multiple GPC values were extracted from different papers, the temperature was averaged and error bars represent the standard deviation across studies.

quantitative cross-study comparisons. However, such metadata are not always systematically reported in ALD publications, underscoring the need for clearer reporting standards and community-wide standardization practices. Incorporating more comprehensive and standardized metadata in future extraction workflows would further enhance the reliability and interpretability of ALD datasets derived from the literature.

## V. CONCLUSION AND FUTURE WORK

In this study, we presented *SciKGExtract*, an agentic AI workflow for structured knowledge extraction from scientific literature, demonstrated on ALD processes. The proposed framework integrates LLMs with deterministic validation, normalization, and cleaning modules, ensuring that extracted information is not only syntactically consistent but also semantically grounded. By leveraging schema-guided extraction, our workflow produces machine-readable knowledge representations that adhere to domain-specific process structures. The incorporation of PubChem-based normalization further enhances consistency by aligning chemically equivalent expressions through PubChem identifiers.

Experimental evaluations on ZnO and *IGZO* ALD datasets revealed that *SciKGExtract* effectively extracts and organizes process information, achieving high semantic similarity and normalization accuracy when compared to expert-annotated ground truth. The results demonstrate that coupling LLMs with agentic tool orchestration significantly improves extraction robustness, mitigates hallucinations, and supports downstream scientific knowledge integration.

While the current workflow focuses on text-based extraction, some valuable scientific information also resides in nontext modalities, such as figures and charts. As part of our future work, we aim to extend *SciKGExtract* by integrating vision-language models (VLMs) to enable multimodal knowledge extraction, thereby broadening the scope of structured data acquisition.[47] The current schema is primarily designed around conventional ALD chemistries based on precursor/coreactant pairs and supercycle-based approaches, and future work will aim to extend its applicability to more complex ALD processes, including ABC-type processes. Additionally, we plan to incorporate a validation layer based on the "LLMs-as-a-judge" paradigm, wherein a secondary LLM critically evaluates and verifies the correctness of extracted information against the original scientific source. This enhancement will further strengthen the workflow's reliability and reduce extraction errors.[48–50]

## SUPPLEMENTARY MATERIAL

See the supplementary material for the SciKGExtract extraction prompt used for information retrieval and list of PubChem Compound Identifiers (CIDs) assigned to the ALD ZnO and *IGZO* precursors analyzed in this work.

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## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

S.S. and E.P. contributed equally to this work.

**Sameer Sadruddin:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (lead); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Eleni Poupaki:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Visualization (lead); Writing – original draft (equal); Writing – review & editing (equal). **Jennifer D'Souza:** Conceptualization (lead); Formal analysis (equal); Funding acquisition (equal); Methodology (equal); Project administration (equal); Resources (equal); Software (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Sören Auer:** Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Resources (lead); Writing – review & editing (equal). **Alex Watkins:** Conceptualization (equal);

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Validation (equal); Writing – review & editing (equal). **Bora Karasulu:** Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Writing – review & editing (equal). **Adriaan J. M. Mackus:** Conceptualization (equal); Funding acquisition (equal); Project administration (equal); Writing – review & editing (equal). **Erwin Kessels:** Conceptualization (equal); Funding acquisition (equal); Project administration (lead); Writing – review & editing (equal).

## DATA AVAILABILITY

The data that support the findings of this study are openly available in public GitHub repository at https://github.com/sciknoworg/scikgextract, Zenodo, https://doi.org/10.5281/zenodo.19714380. Additional methodological details and the PubChem CIDs are provided in the supplementary material.

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